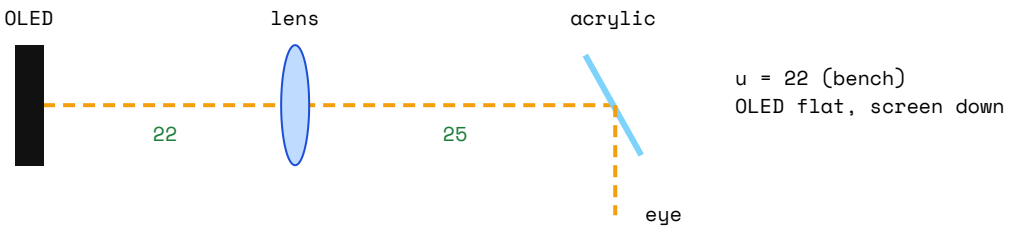
Step 12: measure before 3D printing

HUD glasses · Charan · fill in by hand · all sizes in mm
Tools: ruler, calipers, phone camera, t3_oled_hello uploaded on USB

A. Optics bench (do in order) — no mirror



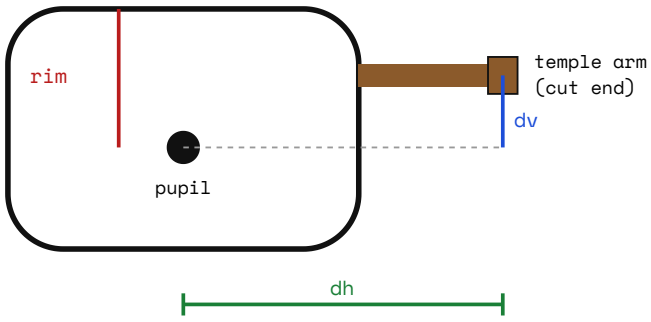
Light path drawn straight. Distances in mm.

1. Hold the lens in front of the OLED. Slide the OLED until HELLO is sharpest.
   Measure OLED screen → lens centre: **u =** _____ mm (bench: 22)
2. Put the acrylic at 45° about **25 mm** behind the lens. Still sharp? □ yes □ no
3. Text reads the right way round with MIRROR_X = □ 0 □ 1 (flip, upload, look again)
4. Readable in normal room light? □ yes □ no (if no: raise contrast, dark paper behind acrylic)
5. Lens real focus: project a bright window onto paper, sharp image → lens to paper **f =** _____ mm

B. Parts (calipers)

Part	Measure	Value (mm)
Acrylic	length × width × thickness width must be 29–30.4	
Lens	diameter, centre thickness	
OLED	board width × length × thickness glass thickness where the glass starts and ends along the board (from the edge without pins)	
TP4056	length × width × tallest part	
ESP32-C3	length × width × board thickness	
Slide switch	body length × width × height	
LiPo	length × width × thickness	

C. Your glasses (wear them · ruler + selfie)



Your right eye as seen in a selfie. Hold a ruler flat next to your face so you can scale the photo.

Measure	Value (mm)
Temple arm width × height (~2 cm behind the hinge)	
dv: pupil → arm centre, up/down (+ = above pupil)	
dh: pupil → arm centre, sideways	
rim: frame top edge above pupil	

Check: rim must be under 26 mm, otherwise the frame hits the bottom of the housing tube.

D. When done

1. Take a photo of this sheet and send it.
2. The 3D model gets rebuilt with your numbers.
3. Print test_fit.stl first, check fits, then the full set.

Pass: text sharp and right way round through lens + acrylic, and every box above filled in.